SEMESTER 2 • BASIC SCIENCE • PHYSICS HANDBOOK

Applied Physics-II

A complete student textbook for wave motion, optics, electromagnetism and modern physics with formulas, diagrams, animations, solved numerical problems, expected questions, model paper and full answers.

COURSE CODE
2003

COURSE TITLE
Applied Physics-II

PROGRAMME
Diploma in Engineering and Technology

SEMESTER
2

CREDITS
3

CATEGORY
Basic Science

PERIODS / WEEK
3 (L:3, T:0, P:0)

PERIODS / SEMESTER
45

45

Periods of physics practice

Learn principle, formula, diagram and numerical method together. Physics becomes easy when concept and calculation are studied as one unit.

Waves | Optics | Electricity | Magnetism | Semiconductor | LASER

COURSE OVERVIEW

What Applied Physics-II is about

Applied Physics-II connects fundamental physics with engineering applications. The course begins with wave motion and sound, moves to lenses and optical fiber, then studies electricity, circuits, magnetism and measuring instruments, and ends with semiconductor devices, photoelectric effect, solar cells, LASERs and nanotechnology.

Engineering applications

Waves explain sound, vibration and ultrasonic testing. Optics explains lenses, microscopes and fiber communication. Electromagnetism explains circuits and meters. Modern physics explains diodes, transistors, solar cells and lasers.

Numerical skill

You will calculate wave velocity, frequency, lens power, magnification, resistance, meter bridge balance and galvanometer conversion.

> Formula □□□□□□ □□□□□□□□□ □□□□. Given data, substitution, unit □□□□□□ □□□□□□□□ □□□□□□.

Concept + diagram

Many physics answers become clear with a neat diagram: convex lens, optical fiber, circuit, galvanometer, p-n junction and laser energy levels.

HOW TO USE THIS HANDBOOK

Best study method

For every topic, follow this order: definition → physical meaning → formula → diagram → solved example → practice problem → revision.

1. Learn concept

Read the English explanation and use Malayalam support only for difficult ideas.

2. Memorise formulas

Understand symbols and units before using formulas.

3. Practise numericals

Write given data, formula, substitution, calculation and final unit.

4. Revise diagrams

Practise drawing labelled diagrams quickly and neatly.

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COURSE OBJECTIVES AND OUTCOMES

Official course direction

Course Objectives

1. To provide students with a broad understanding of physical principles of the universe to help them develop critical thinking and quantitative reasoning skills.
2. To help diploma engineers apply the basic concepts of physics to solve broad-based engineering problems.

Prerequisite

Basic knowledge in Physics at Secondary School level.

You should know basic measurement, units, motion, light, electricity and simple formulas.

> Secondary school physics □□□□□□□□□□□□ □ subject-□ numerical problems easier □□□□□□□□□□.

Course Outcomes

CO	Outcome	Duration	Level	Student meaning
CO1	Calculate the characteristics of waves.	10 h	Applying	Find frequency, wavelength, velocity, beat frequency and acoustics basics.
CO2	Compute the power of lens.	11 h	Applying	Use lens formula, magnification and power of lens.
CO3	Convert galvanometer into ammeter and voltmeter.	11 h	Applying	Use electricity, magnetism and resistance ideas to design meter circuits.
CO4	Explain semiconductor physics, photoelectric effect, LASER action and nanoscience.	11 h	Understanding	Understand diodes, transistors, solar cells and lasers.
Series Test	Series Test	2 h	-	Series Test I after Module 2; Series Test II after Module 4.

CO-PO Mapping Summary

CO1 → PO1

3, strongly mapped.

CO2 → PO1

3, strongly mapped.

CO3 → PO1

3, strongly mapped.

CO4 → PO1

2, moderately mapped.

Mapping scale: 3 - Strongly mapped, 2 - Moderately mapped, 1 - Weakly mapped.

MODULE-WISE LEARNING ROADMAP

45-period plan

Module	Duration	Main topics	Numerical focus	Diagram focus
1. Wave Motion and Its Applications	10 h	SHM, waves, beats, ultrasonics, acoustics	$v=f\lambda$, $T=1/f$, beat frequency	SHM projection, transverse/longitudinal waves
2. Optics	11 h + Series Test I	Reflection, refraction, lenses, optical instruments, TIR, fiber	Lens formula, power, magnification, critical angle	Convex lens, TIR, optical fiber
3. Electromagnetism	11 h	Charge, field, current, resistance, Kirchhoff, bridge, magnetic effects, meters	Ohm's law, resistance, meter bridge, ammeter/voltmeter	Circuit, bridge, galvanometer, meter conversion

4. Modern Physics	11 h + Series Test II	Semiconductors, photoelectric effect, solar cell, lasers, nanoscience	Photon energy if required	Energy bands, p-n junction, diode V-I, laser levels

MODULE 1 • CO1

Wave Motion and Its Applications

Official focus: periodic motion, SHM, projection of uniform circular motion, wave characteristics, superposition, beats, ultrasonic waves, numerical problems and acoustics of buildings.

Module Outcomes

- M1.01 Discuss SHM and properties.
- M1.02 Calculate wave characteristics.
- M1.03 Define ultrasonic waves and list applications.
- M1.04 Understand acoustics of buildings.

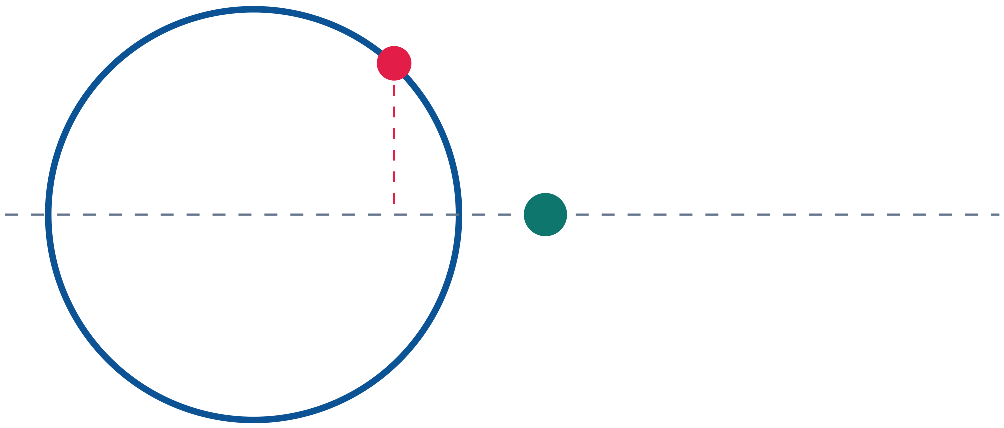
Periodic motion and SHM

Periodic motion repeats after equal intervals of time. Simple Harmonic Motion is a special periodic motion in which acceleration is directly proportional to displacement and always directed towards mean position.

> □□□ □□□□□ □□□□□□□□ motion □□□□□□□□ □□□□□□□□□□□□□ periodic motion. SHM-□ acceleration □□□□□□□□□ mean position-□□□□□□□ □□□□□□□□□□□.

SHM as projection of circular motion

If a particle moves uniformly in a circle, the projection of the particle on any diameter performs SHM. This explains displacement, velocity and acceleration equations.



Projection on diameter performs SHM

SHM formulas

$x = A \sin(\omega t)$

$v = A\omega \cos(\omega t)$

$a = -\omega^2 x$

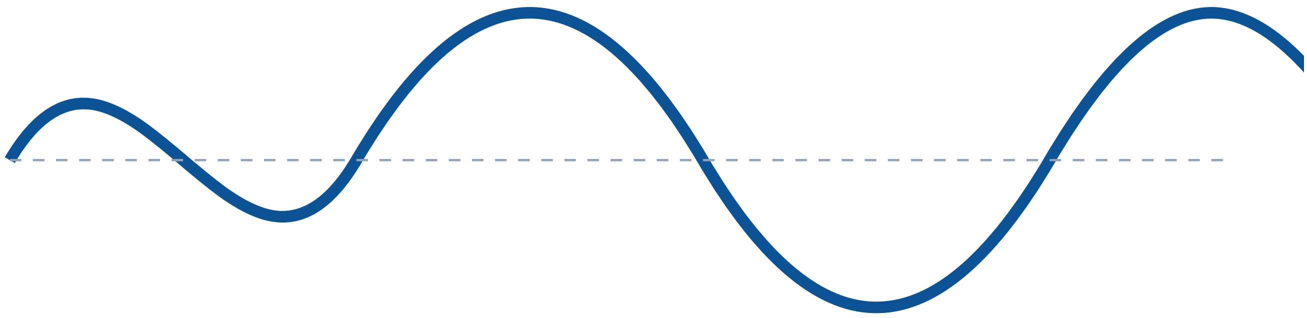
$T = 1/f,\ \ \omega = 2\pi f = 2\pi/T$

Wave motion basics

A wave transfers energy from one place to another without transferring matter as a whole. In mechanical waves, particles vibrate about mean positions.

Transverse wave

Particles vibrate perpendicular to direction of propagation. Examples: string wave, light wave.



Transverse wave

Longitudinal wave

Particles vibrate parallel to direction of propagation. Example: sound wave in air.

Compressions and rarefactions

Wave characteristics

Quantity	Meaning	Unit
Amplitude A	Maximum displacement	m
Wavelength λ	Distance between points in same phase	m
Frequency f	Oscillations per second	Hz
Time period T	Time for one oscillation	s
Wave velocity v	Distance travelled per second	m/s
Phase	State of vibration	radian/degree

$$v = f\lambda$$

Superposition and beats

When waves meet, resultant displacement is algebraic sum of displacements. Beats are periodic loud-soft variations produced by two waves of nearly equal frequency.

$$\text{Beat frequency} = |f_1 - f_2|$$

> Frequency difference □□□□□□□□□□□ sound loud-soft □□□ □□□□□□□□□. □□□□□□ beats.

Ultrasonic waves and acoustics

Ultrasonic waves have frequency above 20 kHz. They are used in crack detection, thickness measurement, ultrasonic cleaning, medical scanning and welding. Acoustics of buildings studies echo, reverberation, noise and methods to improve sound clarity in halls.

Echo

Distinct reflected sound heard after original sound.

Reverberation

Persistence of sound due to multiple reflections.

Control

Use curtains, carpets, acoustic panels, false ceiling and avoid flat reflecting surfaces.

Solved examples

Wave velocity

$f = 500$ Hz, $\lambda = 0.68$ m. $v = f\lambda = 500 \times 0.68 = 340$ m/s.

Beat frequency

f_1 = 256 Hz, f_2 = 260 Hz. Beat frequency = |260 - 256| = 4 Hz.

MODULE 2 • CO2

Optics

Official focus: reflection, refraction, refractive index, lenses, convex lens image formation, lens formula, power, magnification, aberrations, microscope, telescope, TIR and optical fiber.

Module Outcomes

- M2.01 Explain laws of optics and convex lens image formation.
- M2.02 Apply distance relation and discuss lens defects.
- M2.03 Explain optical instruments.
- M2.04 Describe TIR and optical fiber.

Reflection and refraction

Reflection is bouncing back of light. Refraction is bending of light when it passes between media due to speed change.

> Light □□□ medium-□ □□□□□□□ □□□□□□□□□□□□□□□□□ □□□□□□□□□□ speed □□□□□□□□□□□□□ direction □□□□□. □□□□□ refraction.

Laws and refractive index

Reflection

1. Incident ray, reflected ray and normal lie in same plane.
2. Angle of incidence equals angle of reflection.

Refraction

1. Incident ray, refracted ray and normal lie in same plane.
2. sin i / sin r is constant.

$$n = c/v$$

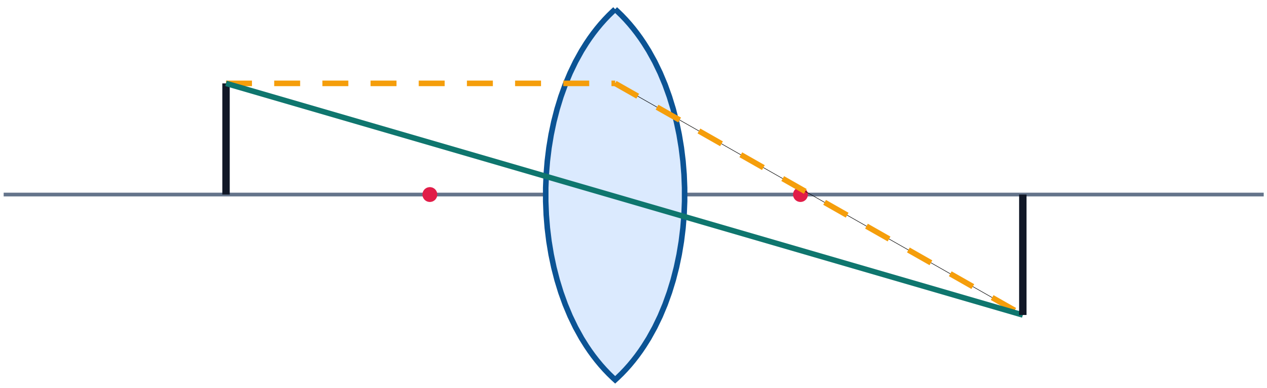
Lenses and formulas

Convex lens

Thicker at centre, converges rays, can form real or virtual images.

Concave lens

Thinner at centre, diverges rays and generally gives virtual erect diminished image.



Convex lens image formation

Lens formula: $1/f = 1/v - 1/u$

Power: $P = 1/f$, where f is in metre. Unit: dioptre.

Magnification: $m = v/u$ = image height/object height

Exam Tip: Convert focal length from centimetre to metre before calculating power.

Lens defects and instruments

Chromatic aberration

Different colours focus at different points, causing coloured fringes.

Spherical aberration

Rays through different parts of spherical lens do not meet at one focus.

Simple microscope

Single convex lens of short focal length producing magnified virtual image.

Astronomical telescope

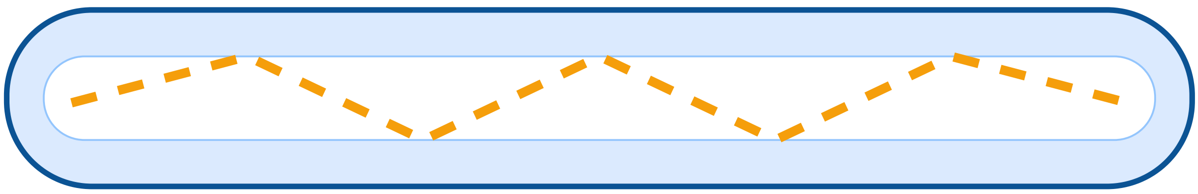
Objective forms real image of distant object and eyepiece magnifies it.

Total internal reflection and optical fiber

TIR occurs when light travels from denser to rarer medium with angle of incidence greater than critical angle. Optical fiber guides light by repeated TIR through core.

$$\sin C = 1/n$$

Light guided by total internal reflection



Solved examples

Power of lens

f = 25 cm = 0.25 m. P = 1/f = 1/0.25 = +4 D.

Lens formula

$u = -30$ cm, $f = +20$ cm. $1/f = 1/v - 1/u$. $1/20 = 1/v + 1/30$. $1/v = 1/20 - 1/30 = 1/60$. $v = 60$ cm.

MODULE 3 • CO3

Electromagnetism

Official focus: Coulomb law, electric field, potential, capacitance, current, Ohm law, resistance, conductance, resistor combinations, colour coding, Kirchhoff laws, bridges, magnetic field, induction, Lorentz force, moving coil galvanometer and conversion.

Module Outcomes

- M3.01 Explain Coulomb law, electric field and potential.
- M3.02 Apply Ohm law and resistance combinations.
- M3.03 Use Kirchhoff laws and meter bridge.
- M3.04 Explain magnetic effect and galvanometer conversion.

Electric charge and field

Charge is a basic property causing electric force. Electric field is the region around a charge where another charge experiences force.

> Charge □□□□□□□□□□□ □□□□□□□□□ □□□□□□□□ charge force □□□□□□□□□□□□. □ region □□□ electric field.

Coulomb law

$$F = (1/4\pi\varepsilon_0)\, q_1 q_2/r^2$$

Field, potential, capacitance

$$E = F/q,\ V = W/q,\ C = Q/V$$

Ohm law and resistance

Electric current is rate of flow of charge. DC flows in one direction; AC changes direction periodically.

I = Q/t, V = IR

R = ρl/A, G = 1/R

Wire □□□□ □□□□□□□□□□ resistance □□□□□□. Area □□□□□□□□□□ resistance □□□□□□□.

Series

$Rs = R_1 + R_2 + R_3 + ...$

Parallel

$1/Rp = 1/R_1 + 1/R_2 + ...$

Colour coding

Digits: Black 0, Brown 1, Red 2, Orange 3, Yellow 4, Green 5, Blue 6, Violet 7, Grey 8, White 9. First two bands give digits, third gives multiplier, fourth gives tolerance.

Kirchhoff, bridge and meter bridge

KCL

Total current entering a junction equals current leaving.

$\Sigma I = 0$

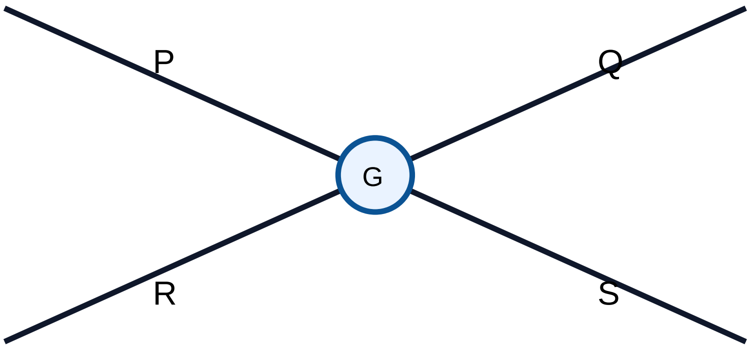
KVL

Algebraic sum of potential differences in a closed loop is zero.

$\Sigma V = 0$

Wheatstone balance: P/Q = R/S

Meter bridge: $R/S = l/(100-l)$



Wheatstone bridge

Magnetism and galvanometer

Faraday law: $\varepsilon = -N\, d\Phi/dt$

Lorentz force: $F = qvB \sin\theta$

Current conductor: $F = BIL \sin\theta$

A moving coil galvanometer detects small current. It becomes an ammeter by a low resistance shunt in parallel and a voltmeter by a high resistance in series.

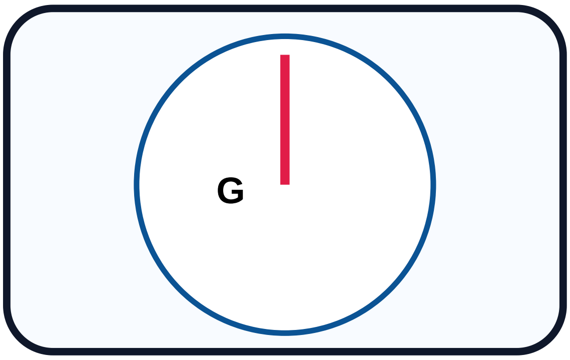
> Ammeter □□□□□□□ parallel shunt, Voltmeter □□□□□□□ series high resistance.

Ammeter conversion

$S = I_gG/(I - I_g)$

Voltmeter conversion

$R = V/I_g - G$



Moving coil galvanometer

Solved examples

Ohm law

R = 6 Ω, I = 2 A. V = IR = 2 × 6 = 12 V.

Parallel resistance

6 Ω and 3 Ω in parallel: 1/R = 1/6 + 1/3 = 3/6 = 1/2. R = 2 Ω.

Galvanometer to ammeter

G = 99 Ω, Ig = 0.01 A, I = 1 A. S = IgG/(I-Ig) = (0.01×99)/(1-0.01)=0.99/0.99=1 Ω. Connect 1 Ω shunt in parallel.

MODULE 4 • CO4

Modern Physics

Official focus: semiconductor physics, energy bands, intrinsic/extrinsic semiconductors, p-n junction, diode V-I characteristics, transistors, photoelectric effect, photocells, solar cells, lasers, He-Ne and semiconductor laser, nanoscience and nanotechnology.

Module Outcomes

- M4.01 Discuss diodes and transistors.
- M4.02 Explain photoelectric effect and applications.
- M4.03 Explain LASER action, semiconductor laser and He-Ne laser.
- M4.04 Summarize nanoscience.

Energy bands in solids

In solids, electrons occupy energy bands. Conductors have overlapping bands, insulators have large forbidden gap, and semiconductors have small forbidden gap.

> Semiconductor-□ forbidden gap □□□□□□□□□. □□□□□□ doping/temperature □□□□□□ conductivity □□□□□.

Conductor

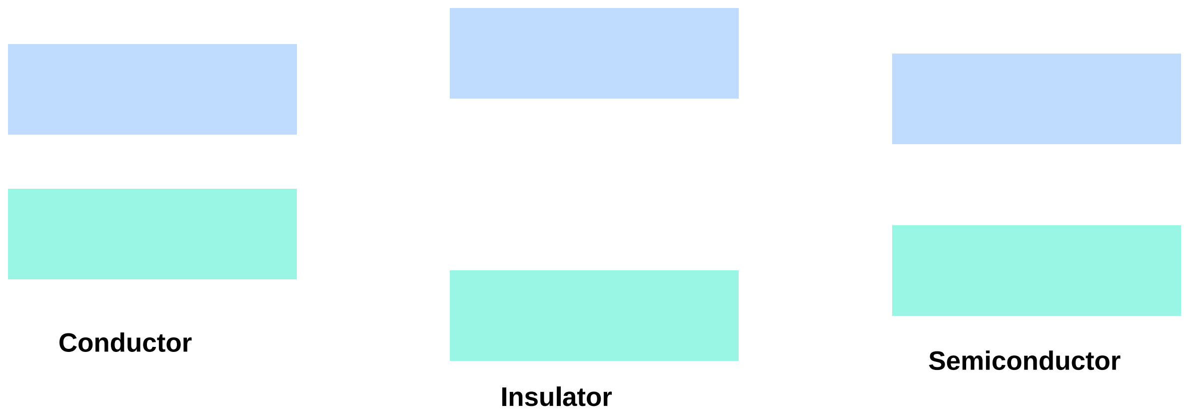
No or very small band gap; many free electrons.

Semiconductor

Small band gap; conductivity can be controlled.

Insulator

Large band gap; very low conductivity.



Semiconductors and p-n junction

Intrinsic

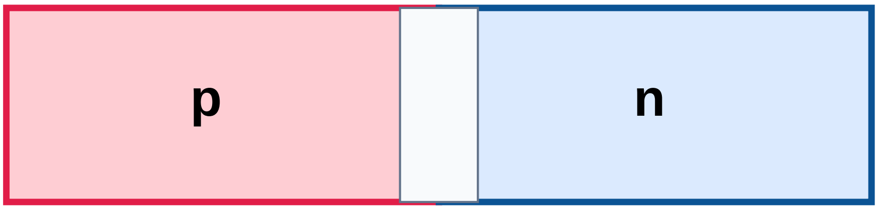
Pure semiconductor with equal electrons and holes.

Extrinsic

Doped semiconductor: n-type has electrons; p-type has holes as majority carriers.

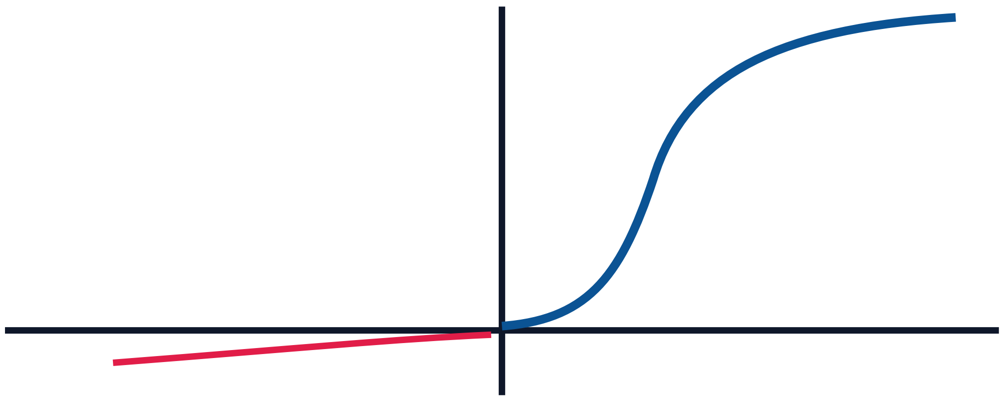
A p-n junction is formed by joining p-type and n-type semiconductors. Forward bias allows current; reverse bias allows only small leakage current.

> Forward bias-□ diode current flow □□□□□□□. Reverse bias-□ current □□□□ □□□□□.



p-n junction

Diode V-I characteristic



Photoelectric effect, solar cell and LASER

Photoelectric effect is emission of electrons when light of suitable frequency falls on a metal. A solar cell converts light energy into electrical energy. LASER means Light Amplification by Stimulated Emission of Radiation.

Photon energy: $E = hf$

Photoelectric equation: $hf = \varphi + KEmax$

$1\ \text{nm} = 10^{-9}\ \text{m}$

> Laser light □□□□□□□ light □□□□ □□□□□□□□□. □□□ coherent, directional, intense □□□.

Laser energy levels

He-Ne laser

Gas laser using helium-neon mixture; common in labs and alignment.

Semiconductor laser

Compact laser diode used in optical communication and devices.

Nanoscience

Study of materials at nanometre scale; properties change due to size.

Applications

Laser cutting, welding, surgery, optical communication; nanotech in sensors, medicine, coatings and energy storage.

FORMULA BANK

Important formulas module-wise

SHM and waves

$x = A \sin(\omega t)$

$v = A\omega \cos(\omega t)$

$a = -\omega^2 x$

$T = 1/f$

$\omega = 2\pi f = 2\pi/T$

$v = f\lambda$

Beat frequency = $|f_1 - f_2|$

Optics

$n = c/v$

$1/f = 1/v - 1/u$

$P = 1/f$, f in metre

$m = v/u = hi/ho$

$\sin C = 1/n$

Electricity

$F = (1/4\pi\varepsilon_0)\, q_1 q_2/r^2$

$E = F/q$

$V = W/q$

$C = Q/V$

$V = IR$

$R = \rho l/A$

$G = 1/R$

$R_s = R_1 + R_2 + \dots$

$1/R_p = 1/R_1 + 1/R_2 + \dots$

Bridge, magnetism, meters

KCL: $\Sigma I = 0$

KVL: $\Sigma V = 0$

$P/Q = R/S$

$R/S = l/(100-l)$

$\varepsilon = -N\, d\Phi/dt$

$F = qvB \sin\theta$

$F = BIL \sin\theta$

$S = IgG/(I-Ig)$

$R = V/Ig - G$

Modern Physics

$E = hf$

$hf = \varphi + KEmax$

$1 \text{ nm} = 10^{-9} \text{ m}$

UNITS AND SYMBOLS BANK

Symbols and SI units

Quantity	Symbol	Unit
Displacement	x	m
Amplitude	A	m
Frequency	f	Hz
Time period	T	s
Wave velocity	v	m/s
Wavelength	λ	m
Refractive index	n	No unit
Focal length	f	m
Power of lens	P	D

Current	I	A
Voltage	V	V
Resistance	R	Ω
Resistivity	ρ	Ω m
Conductance	G	S
Capacitance	C	F
Charge	Q	C
Electric field	E	N/C
Magnetic field	B	T
Energy	E	J
Nanometre	nm	10^{-9} m

IMPORTANT DIAGRAMS BANK

Diagrams to practise

Module 1

- SHM projection
- Transverse wave
- Longitudinal wave
- Beat concept

Module 2

- Convex lens ray diagram
- Total internal reflection
- Optical fiber propagation
- Microscope/telescope

Module 3

- Ohm law circuit
- Series and parallel circuits
- Wheatstone bridge
- Meter bridge
- Moving coil galvanometer
- Ammeter/voltmeter conversion

Module 4

- Energy band diagrams
- p-n junction
- Diode V-I characteristics
- pnp/npn symbols
- Photoelectric setup
- Solar cell
- Laser energy levels

Diagram method

1. Draw large enough.
2. Use neat labels.
3. Show direction of rays/current where needed.
4. Write one explanation below.

SOLVED NUMERICAL EXAMPLES

Calculation practice from all numerical modules

Time period and frequency

50 oscillations in 10 s. f = 50/10 = 5 Hz. T = 1/f = 1/5 = 0.2 s.

Critical angle

n = 1.5. sin C = 1/n = 1/1.5 = 0.6667. C ≈ 41.8°.

Law of resistance

$l=2$ m, $A=1\times10^{-6}$ m², $\rho=1.7\times10^{-8}$ Ωm. $R=\rho l/A=(1.7\times10^{-8}\times2)/(1\times10^{-6})=0.034$ Ω.

Meter bridge

Balance length l=40 cm, S=6 Ω. R/S=l/(100-l). R/6=40/60. R=4 Ω.

Galvanometer to voltmeter

G=50 Ω, Ig=5 mA=0.005 A, V=10 V. R=V/Ig-G=10/0.005-50=2000-50=1950 Ω.

EXPECTED QUESTIONS

Module-wise question bank

Module 1

Very short

1. Define SHM.
2. What is wavelength?
3. Define ultrasonic waves.

Numerical

1. Find wave velocity.
2. Find beat frequency.

Long

1. Explain SHM projection.

2. Explain acoustics defects and remedies.

Module 2

Very short

1. Define refractive index.
2. What is lens power?
3. Define critical angle.

Numerical

1. Find power of lens.
2. Solve lens formula problem.

Long

1. Explain convex lens image formation.
2. Explain optical fiber.

Module 3

Very short

1. State Ohm law.
2. Define capacitance.
3. What is KCL?

Numerical

1. Ohm law.
2. Resistance combination.
3. Galvanometer conversion.

Long

1. Explain moving coil galvanometer.
2. Explain meter bridge.

Module 4

Very short

1. Define semiconductor.

2. What is population inversion?
3. What is 1 nanometre?

Diagram

1. Energy band diagram.
2. Diode V-I curve.
3. Laser energy levels.

Long

1. Explain p-n junction diode.
2. Explain laser action.

Objective / MCQ

No	Question	Options
1	Wave relation is	A $v=f/\lambda$ B $v=f\lambda$ C $f=v\lambda$ D $\lambda=vf$
2	Unit of lens power is	A watt B dioptre C tesla D ohm
3	Series equivalent resistance is	A less than least B product C sum D zero
4	Majority carriers in n-type are	A holes B electrons C photons D protons
5	LASER light is	A incoherent B divergent C coherent D white

PRACTICE PROBLEMS

Module-wise practice with answers

Module 1 numericals

1. v=340 m/s, f=170 Hz. Find λ. **Ans: 2 m**
2. T=0.02 s. Find f. **Ans: 50 Hz**
3. 512 Hz and 520 Hz. Beat? **Ans: 8 Hz**

Module 2 numericals

1. Find P for f=20 cm. **Ans: +5 D**
2. Find f for P=-2 D. **Ans: -0.5 m**

3. Find C for n=1.414. **Ans: 45°**

Module 3 numericals

1. I=3 A, R=4 Ω. Find V. **Ans: 12 V**
2. 2 Ω,3 Ω,5 Ω in series. **Ans: 10 Ω**
3. 4 Ω and 4 Ω in parallel. **Ans: 2 Ω**

Module 4 practice

1. Draw energy bands.
2. Explain forward and reverse bias.
3. Write four LASER applications.

Self-assessment

- ✓ I can write formulas with units.
- ✓ I can solve lens power problems.
- ✓ I can solve resistance problems.
- ✓ I can draw optical fiber diagram.
- ✓ I can explain laser action.

QUICK REVISION

Last-day revision sheet

Module 1

- $a=-\omega^2 x$
- $v=f\lambda$
- Beat=$|f_1-f_2|$
- Reverberation is persistence of sound.

Module 2

- $n=c/v$
- $1/f=1/v-1/u$
- $P=1/f$ metre
- TIR: denser to rarer and $i>C$.

Module 3

- $V=IR$
- $R=\rho l/A$
- KCL and KVL
- Ammeter: shunt, voltmeter: series R

Module 4

- p-n junction
- Photoelectric effect
- Laser needs population inversion
- 1 nm=10^{-9} m

Common Mistakes: using cm instead of metre for lens power, forgetting units, mixing ammeter and voltmeter conversion, and drawing unlabeled diagrams.

MODEL QUESTION PAPER WITH FULL ANSWERS

APPLIED PHYSICS-II • Maximum Marks: 75 • Time: 3 Hours

Fresh model paper based on the syllabus. Pattern: very short, short/numerical, and long/application questions.

PART A - Answer all. [10 × 1 = 10]

1. Define SHM.
2. Write v, f, λ relation.
3. What is reverberation?
4. Define power of lens.
5. State one TIR condition.

6. State Ohm's law.
7. SI unit of capacitance?
8. What is connected to convert galvanometer into ammeter?
9. What is population inversion?
10. Write one nanometre in metre.

PART B - Answer any five. [5 × 5 = 25]

1. Explain SHM as projection of circular motion.
2. A wave has f=250 Hz and λ=1.36 m. Calculate v.
3. Explain chromatic and spherical aberration.
4. A convex lens has f=25 cm. Find power.
5. State Kirchhoff laws and one application.
6. Find equivalent resistance of 4 Ω and 6 Ω in series and parallel.
7. Explain photoelectric effect and one application.

PART C - Answer any four. [4 × 10 = 40]

1. Explain transverse and longitudinal waves; derive v=fλ and solve: f=680 Hz, v=340 m/s. Find λ.
2. Explain TIR and optical fiber with diagram and applications.
3. Describe moving coil galvanometer and conversion into ammeter and voltmeter.
4. G=99 Ω, Ig=0.01 A. Find shunt for 1 A ammeter and explain principle.
5. Explain energy bands and compare conductor, semiconductor and insulator.
6. Explain LASER action with energy level diagram, characteristics and applications.

ANSWER KEY / COMPLETE SOLUTIONS

Full answers for model paper

Part A Answers

No	Answer
1	Periodic motion with acceleration proportional to displacement and directed to mean position.
2	v=fλ.
3	Persistence of sound due to multiple reflections after source stops.
4	Reciprocal of focal length in metre: P=1/f.

5	Light travels denser to rarer medium and i>C.
6	At constant temperature, V is proportional to I; V=IR.
7	Farad.
8	Low resistance shunt in parallel.
9	More atoms in excited state than lower state.
10	10^{-9} m.

Part B Solutions

B1 SHM projection

A particle moving uniformly in a circle has a projection on a diameter. This projection moves to and fro along the diameter and performs SHM. If A is radius and ω is angular velocity, displacement $x=A \sin\omega t$. Acceleration is directed towards mean position. Malayalam: Circular motion-□□□□ projection □□□□□□□□□□□□ oscillation □□□□ □□□□□□□□□□□□ SHM.

B2 Wave velocity

$f=250$ Hz, $\lambda=1.36$ m. $v=f\lambda=250\times1.36=340$ m/s.

B3 Lens defects

Chromatic aberration: different colours focus at different points, producing coloured fringes. Spherical aberration: rays through different parts of a spherical lens do not meet at one focus, producing blur.

B4 Power

$f=25$ cm$=0.25$ m. $P=1/f=1/0.25=+4$ D.

B5 Kirchhoff laws

KCL: Total current entering a junction equals total current leaving. KVL: Algebraic sum of potential differences in a closed loop is zero. Applications: circuit analysis, Wheatstone bridge and meter bridge.

B6 Resistance

Series: $R_s=4+6=10\ \Omega$. Parallel: $1/R_p=1/4+1/6=5/12$; $R_p=12/5=2.4\ \Omega$.

B7 Photoelectric effect

Photoelectric effect is emission of electrons from a metal surface when light of suitable frequency falls on it. It proves particle nature of light. Applications: photocell and solar cell.

Part C Solutions

C1 Waves

Transverse wave: particle vibration perpendicular to direction of propagation. Example: light wave. Longitudinal wave: particle vibration parallel to direction of propagation. Example: sound in air. In one time period T, wave travels one wavelength λ. Hence $v=\lambda/T=f\lambda$. Numerical: $v=340$ m/s, $f=680$ Hz. $\lambda=v/f=340/680=0.5$ m.

C2 TIR and optical fiber

TIR occurs when light travels from denser to rarer medium and angle of incidence is greater than critical angle. In optical fiber, core has higher refractive index than cladding, so light undergoes repeated TIR and travels through core. Applications: telecommunication, internet, endoscopy and sensors. Malayalam: Fiber-□ light core-□ reflect □□□□□□ □□□□□□□□□□ □□□□□□□□□□□□ signal long distance □□□□□.

C3 Galvanometer conversion

Moving coil galvanometer detects small current by deflection of a coil in magnetic field. Ammeter conversion: connect low resistance shunt in parallel; $S=I_gG/(I-I_g)$. Voltmeter conversion: connect high resistance in series; $R=V/I_g-G$. Malayalam: Ammeter □□□□□□ shunt parallel; voltmeter □□□□□□ high resistance series.

C4 Ammeter numerical

G=99 Ω, Ig=0.01 A, I=1 A. S=IgG/(I-Ig)=(0.01×99)/(1-0.01)=0.99/0.99=1 Ω. Final answer: connect 1 Ω shunt in parallel.

C5 Energy bands

In solids, electron energy levels form bands. Valence band and conduction band are separated by forbidden gap. Conductors have overlapping bands or no gap. Insulators have large gap. Semiconductors have small gap; doping changes their conductivity.

C6 LASER action

LASER works by stimulated emission. Pumping creates population inversion. An incident photon stimulates an excited atom to emit another identical photon. This produces coherent, monochromatic, directional and intense light. Applications: cutting, welding, optical communication, eye surgery and medical treatment.

MCQ Answer Key

1. B - v=fλ. 2. B - dioptre. 3. C - sum. 4. B - electrons. 5. C - coherent.

REFERENCES

Text / Reference books from syllabus

Code	Book Title / Author
T1	Text Book of Physics for Class XI & XII, Part-I and Part-II, N.C.E.R.T., Delhi
R2	Applied Physics, Vol. I and Vol. II, TTTI Publications, Tata McGraw Hill, Delhi
R3	Concepts of Physics by H. C. Verma, Vol. I and II, Bharti Bhawan Ltd., New Delhi
R4	Fundamentals of Physics, Halliday / Resnick / Walker, Wiley India Pvt. Ltd.

R5 Modern Approach to Applied Physics-I and II, A. S. Vasudeva, Modern Publishers

FINAL EXAM REVISION CHECKLIST

Before the exam

Concept checklist

- ✓ I can define SHM, TIR, capacitance, photoelectric effect and population inversion.
- ✓ I can explain optical fiber, galvanometer, p-n junction and laser action.
- ✓ I can draw important diagrams.

Numerical checklist

- ✓ I can use $v=f\lambda$ and beat frequency.
- ✓ I can solve lens power and lens formula.
- ✓ I can solve resistance combinations and meter bridge.
- ✓ I can solve galvanometer conversion.
- ✓ I always write units.